ASSIGNMENT-4

1. Create a new GitHub repository.

- **Clone the repository to your local machine using SSH (generate an SSH key if needed, add the public key to your GitHub account).**
- **Create a new branch named after your username (e.g., Tutedude).**
- **Add your Flask project files to this branch.**
- **Commit the changes and merge the branch into the main branch.**

```
C:\Windows\System32\cmd.e

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout -b himanshu
Switched to a new branch 'himanshu'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git branch

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added Flask project files"
[himanshu (root-commit) 9a59268] Added Flask project files
 6 files changed, 136 insertions(+)
 create mode 100644 flask_mongo_api/.env
 create mode 100644 flask_mongo_api/app.py
 create mode 100644 flask_mongo_form/.env
 create mode 100644 flask_mongo_form/app.py
 create mode 100644 flask_mongo_form/templates/form.html
 create mode 100644 flask_mongo_form/templates/success.html

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin himanshu
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 1.89 KiB | 138.00 KiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:HimanshuUyt/flask-project.git
 * [new branch]      himanshu -> himanshu

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
error: pathspec 'main' did not match any file(s) known to git

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
fatal: couldn't find remote ref main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge himanshu
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'github.com:HimanshuUyt/flask-project.git'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git branch -M main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
```

```
C:\Windows\System32\cmd.e

fatal: couldn't find remote ref main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge himanshu
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'github.com:HimanshuUyt/flask-project.git'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git branch -M main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Already on 'main'
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
fatal: couldn't find remote ref main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge himanshu
merge: himanshu - not something we can merge

Did you mean this?
        origin/himanshu

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:      https://github.com/HimanshuUyt/flask-project/pull/new/main
remote:
To github.com:HimanshuUyt/flask-project.git
 * [new branch]      main -> main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>
```

2)

- **Update the content of the JSON file used for the /api route in this branch.**
- **Merge the <your_name>_new branch into the main branch.**
- **If there are conflicts during the merge, resolve them by accepting the changes from the <your_name>_new branch.**
- **Add the resolved changes to the staging area, commit them, and push the updates to the remote repository.**

```
C:\Windows\System32\cmd.e

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git branch
* main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout -b himanshu_new
Switched to a new branch 'himanshu_new'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Updated JSON data for API"
On branch himanshu_new
nothing to commit, working tree clean

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin himanshu_new
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'himanshu_new' on GitHub by visiting:
remote:      https://github.com/HimanshuUyt/flask-project/pull/new/himanshu_new
remote:
To github.com:HimanshuUyt/flask-project.git
 * [new branch]      himanshu_new -> himanshu_new

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
From github.com:HimanshuUyt/flask-project
 * branch            main       -> FETCH_HEAD
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge himanshu_new
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Resolved merge conflict using himanshu_new changes"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>
```

3)

- **Create two branches: master_1 and master_2 from the main branch.**
- **Feature Development in master_1:**
- **In the master_1 branch, create a To-Do Page in the frontend.**
  - **The page should contain a form with the following fields:**
    - **Item Name**
    - **Item Description**
- **Backend API in master_2:**
- **In the master_2 branch, create a backend route named /submittodoitem.**
- **This route will:**
  - **Accept itemName and itemDescription via a POST request.**
  - **Store these details in a MongoDB database.**
- **Merging Changes:**
- **Merge the changes from both master_1 and master_2 into the main branch.**

```
C:\Windows\System32\cmd.e

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
From github.com:HimanshuUyt/flask-project
 * branch            main       -> FETCH_HEAD
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout -b master_1
Switched to a new branch 'master_1'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout -b master_2
Switched to a new branch 'master_2'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git branch
  himanshu_new
  main
  master_1
* master_2

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout master_1
Switched to branch 'master_1'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added ToDo frontend page in master_1"
[master_1 2d16410] Added ToDo frontend page in master_1
 2 files changed, 26 insertions(+)
 create mode 100644 flask_mongo_form/templates/todo.html

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin master_1
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 814 bytes | 271.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
```

```
C:\Windows\System32\cmd.e

remote:
remote: Create a pull request for 'master_1' on GitHub by visiting:
remote:      https://github.com/HimanshuUyt/flask-project/pull/new/master_1
remote:
To github.com:HimanshuUyt/flask-project.git
 * [new branch]      master_1 -> master_1

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout master_2
Switched to branch 'master_2'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added backend API /submittodoitem"
[master_2 3263d54] Added backend API /submittodoitem
 1 file changed, 14 insertions(+), 37 deletions(-)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin master_2
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 719 bytes | 143.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master_2' on GitHub by visiting:
remote:      https://github.com/HimanshuUyt/flask-project/pull/new/master_2
remote:
To github.com:HimanshuUyt/flask-project.git
 * [new branch]      master_2 -> master_2

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
From github.com:HimanshuUyt/flask-project
 * branch            main       -> FETCH_HEAD
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge master_1
Updating 9a59268..2d16410
Fast-forward
 flask_mongo_form/app.py              |  3 +++
 flask_mongo_form/templates/todo.html | 23 ++++++++++++++++++++++
```

```
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Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git pull origin main
From github.com:HimanshuUyt/flask-project
 * branch            main       -> FETCH_HEAD
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge master_1
Updating 9a59268..2d16410
Fast-forward
 flask_mongo_form/app.py               |  3 +++
 flask_mongo_form/templates/todo.html | 23 +++++++++++++++++++++++
 2 files changed, 26 insertions(+)
 create mode 100644 flask_mongo_form/templates/todo.html

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge master_2
Auto-merging flask_mongo_form/app.py
Merge made by the 'ort' strategy.
 flask_mongo_form/app.py | 51 +++++++++++++----------------------------------------
 1 file changed, 14 insertions(+), 37 deletions(-)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge master_2
Already up to date.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 523 bytes | 261.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:HimanshuUyt/flask-project.git
   9a59268..988d74b  main -> main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>
```

4. **Enhancing the To-Do Form in** master_1:
   - In the master_1 branch, add the following fields to the To-Do form:
     - **Item ID**
     - **Item UUID**
     - **Item Hash**
   - **Committing in Sequence**:
   - Add and commit each field separately in the following order:
     - **First commit**: Add **Item ID** field.
     - **Second commit**: Add **Item UUID** field.
     - **Third commit**: Add **Item Hash** field.
   - **Merging to** main:
   - Merge the master_1 branch into the main branch.
   - **Git Reset and Commit Deletion**:
   - In the main branch, use **Git Reset** to roll back to the commit where only the **Item ID** field was added.
   - Use git reset --soft to ensure changes remain staged.
   - Re-commit this state to the main branch.
   - Merge this updated state to the main branch.
   - **Rebasing Changes**:
   - Rebase the updated changes in the main branch to the master_1 branch.
     Clarification:
     - During rebasing, **preserve individual commits** to maintain the commit history for each change (i.e., do not squash commits).
     - Use git rebase main master_1 to integrate changes from the main branch back into the master_1 branch.

```
C:\Windows\System32\cmd.e

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout master_1
Switched to branch 'master_1'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added Item ID field"
[master_1 2b801f4] Added Item ID field
 1 file changed, 3 insertions(+)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added Item UUID field"
[master_1 de5ee89] Added Item UUID field
 1 file changed, 3 insertions(+)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Added Item Hash field"
[master_1 b08e857] Added Item Hash field
 1 file changed, 3 insertions(+)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git merge master_1
Merge made by the 'ort' strategy.
 flask_mongo_form/templates/todo.html | 9 +++++++++
 1 file changed, 9 insertions(+)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main
Enumerating objects: 26, done.
Counting objects: 100% (24/24), done.
Delta compression using up to 8 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (18/18), 1.51 KiB | 154.00 KiB/s, done.
Total 18 (delta 10), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (10/10), completed with 4 local objects.
To github.com:HimanshuUyt/flask-project.git
   988d74b..d92e9c2  main -> main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git log --oneline
d92e9c2 (HEAD -> main, origin/main) Merge branch 'master_1'
```

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C:\Windows\System32\cmd.e
   988d74b..d92e9c2  main -> main

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git log --oneline
d92e9c2 (HEAD -> main, origin/main) Merge branch 'master_1'
b08e857 (master_1) Added Item Hash field
de5ee89 Added Item UUID field
2b801f4 Added Item ID field
988d74b Merge branch 'master_2'
3263d54 (origin/master_2, master_2) Added backend API /submittodoitem
2d16410 (origin/master_1) Added ToDo frontend page in master_1
9a59268 (origin/himanshu_new, origin/himanshu, himanshu_new) Added Flask project files

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git reset --soft 2b801f4

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git commit -m "Reset main branch to Item ID field state"
[main 7fcbebc] Reset main branch to Item ID field state
 2 files changed, 20 insertions(+), 37 deletions(-)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin main --force
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 888 bytes | 222.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:HimanshuUyt/flask-project.git
 + d92e9c2...7fcbebc main -> main (forced update)

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git checkout master_1
Switched to branch 'master_1'

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git rebase main
Auto-merging flask_mongo_form/templates/todo.html
CONFLICT (content): Merge conflict in flask_mongo_form/templates/todo.html
error: could not apply de5ee89... Added Item UUID field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply de5ee89... # Added Item UUID field

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .
```

```
C:\Windows\System32\cmd.e

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git rebase main
Auto-merging flask_mongo_form/templates/todo.html
CONFLICT (content): Merge conflict in flask_mongo_form/templates/todo.html
error: could not apply de5ee89... Added Item UUID field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply de5ee89... # Added Item UUID field

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git add .

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git rebase --continue
[detached HEAD 7def9ce] Added Item UUID field
 1 file changed, 3 insertions(+)
Auto-merging flask_mongo_form/templates/todo.html
CONFLICT (content): Merge conflict in flask_mongo_form/templates/todo.html
error: could not apply b08e857... Added Item Hash field
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply b08e857... # Added Item Hash field

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>git push origin master_1 --force
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 847 bytes | 105.00 KiB/s, done.
Total 10 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 3 local objects.
To github.com:HimanshuUyt/flask-project.git
   2d16410..b08e857  master_1 -> master_1

D:\TuteDudeCourses\DevOps\Assignment\Assignment 4\flask-project>
```